Bound Entangled States Are Useful in Prepare-and-Measure Scenarios

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We show that bipartite bound entangled states make possible violations of correlation inequalities in the prepare-and-measure scenario. These inequalities are satisfied by all classical models as well as by all quantum models that do not feature entanglement. In contrast to the known Bell inequality violations from bound entangled states, we find that the violations in the prepare-and-measure scenario are sizeable and significantly noise-tolerant. Furthermore, we show that significantly stronger quantum correlations are made possible by considering bound entanglement with a larger dimension.

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Introduction—Bound entanglement is a fascinating phenomenon in the theory of quantum entanglement. These are entangled states that, even when available in infinitely many copies, cannot be transformed to a maximally entangled state via local operations and classical communication [1]. In other words, these states are not useful for entanglement distillation. In bipartite systems, it is standard to detect them by verifying that an entangled state has positive partial transpose (PPT). Whether there exists also bound entanglement that is not PPT is a famous open problem [2].

Motivated by understanding the fundamentals of quantum theory, it has for long been asked whether bound entangled states can give rise to various quantum phenomena. For instance, they do support quantum metrology [3] but not quantum teleportation [4], albeit some variations of the task are possible for every entangled state [5,6]. A more fundamental question is whether bound entanglement can violate principles of classicality in black-box correlation experiments. For example, when entangled states are available in many copies, distillability is necessary and sufficient for violating the Clauser-Horne-Shimony-Holt Bell inequality [7]. In 1999 it was conjectured that bound entangled states cannot violate any Bell inequality at all [8] but this was disproved 15 years later through a counterexample based on two qutrits [9]. The reported violation is strikingly small; already an isotropic noise rate on fractions of a per mille is sufficient to render the experiment classical. Also subsequent works have found only small violations of Bell tests with bound entanglement [10–13]. This suggests that the original conjecture may have been "almost true."

A complementary line of research focuses on the prepare-and-measure scenario. Here, one is interested in the correlations created between Alice and Bob when they share entanglement and communicate messages [14,15]. If the message is classical, it is well-known that entanglement enables nonclassical correlations [16] but also that this advantage fundamentally is propelled by Bell nonlocality [17,18]. Consequently, if bound entanglement is a resource, it remains limited by the need for significant Bell inequality violations. However, the situation changes when the messages themselves can be quantum states. This is famously exemplified by the dense coding protocol [19]. In dense coding, qubit messages have no advantage over classical bit messages unless shared entanglement is also made available. Many entangled states that violate no Bell inequality can be used for a quantum advantage in dense coding [20,21], but it turns out that bound entangled states are not among these [22].

Here, we show that bound entangled states of two particles make possible quantum correlations in the prepare-and-measure scenario that significantly violate classical limitations. Specifically, we show large violations of correlation inequalities that are satisfied by all classical models and all quantum models that do not exploit entanglement. This shows not only that bound entanglement is responsible for breaking the classical limitations but also that its advantages can be sizable and therefore also of practical relevance. In our analysis, we consider the black-box input-output scenario based on d-dimensional messages recently introduced in [23]. Using semidefinite relaxation techniques [24], we prove a relevant correlation inequality in this scenario. Then, through an explicit construction, we show that it can be violated with bound entangled states of two qutrits. As a benchmark of the violation, the bound entanglement remains nonclassical when exposed to up to 18.8% of isotropic noise, which far

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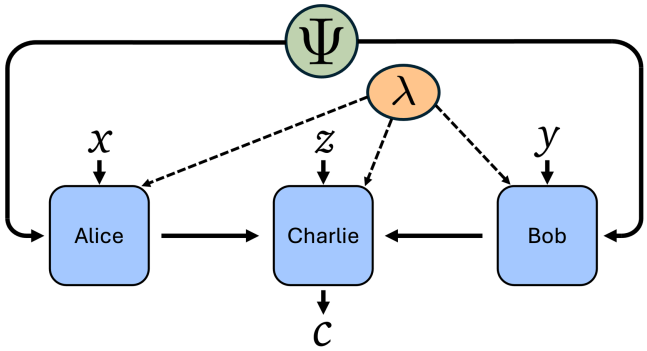

FIG. 1. Symmetric prepare-and-measure scenario. All parties share classical randomness (λ). When entanglement is allowed, any quantum state Ψ may be shared between Alice and Bob. They respectively encode their inputs x and y into d-dimensional messages that are sent to Charlie who decodes them with respect to his input z and outputs c.

exceeds the known counterparts in bipartite Bell scenarios. We subsequently consider correlation inequalities tailored for systems with a higher Hilbert space dimension and prove that even more noise-tolerant quantum correlations are now made possible by using bound entangled states. Our results establish bound entanglement as useful for the prepare-and-measure scenario in both principle and practice.

Scenario—We consider the symmetric prepare-and-measure scenario illustrated in Fig. 1. Alice, Bob, and Charlie select private inputs x, y, and z, respectively. Only Charlie produces an output, and it is labeled c. Alice and Bob can each send a message to Charlie and these messages support at most d distinct symbols. Thus, when the message is a quantum system, it corresponds to a density matrix with Hilbert space dimension d. Moreover, all three parties share unlimited classical randomness, corresponding to a hidden variable, λ, subject to some arbitrary probability distribution $\{q_\lambda\}$. This can be used to stochastically coordinate the communication strategies between the three parties. Also, Alice and Bob can share some arbitrary entangled state, Ψ, to assist their respective selection of messages. The correlations obtained between the three parties, $p(c|x,y,z)$, are in this entanglement-based model given by

$$p_E(c|x,y,z) = \sum_\lambda q_\lambda \mathrm{tr}[(\Lambda_x^\lambda \otimes \Omega_y^\lambda)[\Psi] M_{c|z}^\lambda], \tag{1}$$

where $\{M_{c|z}^\lambda\}$ are the measurements of Charlie, and $\{\Lambda_x^\lambda\}$ and $\{\Omega_y^\lambda\}$ are the message encoding maps of Alice and Bob, respectively, all when conditioned on the hidden variable. Each of the encoding maps corresponds to a completely positive trace-preserving map. These maps transform the arbitrary input space associated with the respective shares of Ψ into a d-dimensional system.

Consider now a classical model of the same scenario. All three parties can still preshare the hidden variable but the entangled state, Ψ, is no longer available. Also, the communication itself is classical, i.e., Alice and Bob map their respective inputs to classical messages $m_A \in [d]$ and $m_B \in [d]$, which are sent to Charlie. We define $[d] \equiv \{0, \ldots, d-1\}$. Charlie then computes his output based on the received messages, his own input z, and the hidden variable λ. Thus, the classical correlations take the form

$$p_C(c|x,y,z) = \sum_{\lambda, m_A, m_B} q_\lambda p(m_A|x,\lambda) p(m_B|y,\lambda) p(c|m_A, m_B, z, \lambda). \tag{2}$$

The space of classical correlations in prepare-and-measure scenarios takes the geometric form of a polytope [25] and its facets correspond to linear correlation inequalities satisfied by all classical models. In general, these inequalities take the form

$$\sum_{c,x,y,z} s_{cxyz} p(c|x,y,z) \leq \beta, \tag{3}$$

for some real coefficients s_{cxyz} and some tight bound β. Thus, violating such an inequality implies nonclassical correlations in the prepare-and-measure scenario. However, a violation is insufficient to guarantee that entanglement is the essential resource behind the failure of classicality. In order to guarantee the role of entanglement, one must also show that the inequality (3) is satisfied also by all quantum models that do not use entanglement, i.e., when the hidden variable is used for coordinating the communication of quantum messages to Charlie. The correlations in such an "unentangled" quantum model take the form

$$p_Q(c|x,y,z) = \sum_\lambda q_\lambda \mathrm{tr}[(\rho_x^\lambda \otimes \sigma_y^\lambda) M_{c|z}^\lambda], \tag{4}$$

where $\{\rho_x^\lambda\}$ and $\{\sigma_y^\lambda\}$ are the d-dimensional quantum messages of Alice and Bob when conditioned on the hidden variable. The set of correlations that admit this model is convex and constitutes, in general, a strict superset of the classical polytope. The latter follows from the fact that quantum messages can lead to nonclassical correlations in the prepare-and-measure scenario [26]. In what follows, we proceed to construct specific correlation inequalities of the form (3), which has the key feature that they are satisfied by both classical models (2), and quantum models without entanglement (4).

Correlation inequality—We consider the input-output scenario introduced in [23]. It is parametrized by d, with $x \equiv (x_0, x_1) \in [d]^2$, $y \equiv (y_0, y_1) \in [d]^2$, $z \in [d+1]$, and $c \in [d]$. Each z corresponds to a game that has a unique right winning answer, $w_z(x,y)$, and Charlie's aim is to output $c = w_z(x,y)$. We define the winning answers as

$$\begin{aligned} z \neq d&: \; w_z = x_1 + y_1 - 2z(x_0 - y_0) \mod d \\ z = d&: \; w_d = x_0 - y_0 \mod d \end{aligned} \tag{5}$$

and define the average winning probability as

$$\mathcal{R}_d = \frac{1}{d^4(d+1)} \sum_{x,y,z} p[c = w_z(x,y)|x,y,z]. \tag{6}$$

While this definition applies to arbitrary d, our focus will be on prime d due to the favourable property of there always existing multiplicative inverses for the winning conditions. This is expected to be useful for finding entanglement-based advantages. It was conjectured in [23] that for any quantum model without shared entanglement and every odd prime d it holds that

$$\mathcal{R}_d \leq \frac{2}{d+1}. \tag{7}$$

It is easy to see that this conjectured bound can be saturated also in a classical model. Let Alice and Bob send $m_A = x_0$ and $m_B = y_0$ to Charlie. When $z = d$, Charlie will correctly answer $c = w_d$ and when $z \neq d$ he can make a random guess that has winning probability $p(c = w_z) = 1/d$. This leads to $\mathcal{R}_d = 2/(d+1)$. Thus, if the conjecture is true, Eq. (7) is an instance of a correlation inequality (3) that is tight for both for classical models and quantum models without entanglement.

We have proven the conjecture for $d = 3, 5, 7$. The proof is based on identifying specific supersets of the set of quantum correlations (4) achievable without entanglement, and then showing that the conjecture (7) is true also for these sets. The supersets are constructed so that they can be characterized as semidefinite programs. Such programs can be solved exactly, and via duality theory they can also provide analytical certificates of the result [24,27].

To this end, we define a tracial moment matrix with elements $\Gamma_{u,v} = \mathrm{tr}(u^\dagger v)$ where $u, v \in L_k$ are monomials up to a selected degree k over the operator list $\mathcal{S} = \{\Pi_A \otimes \Pi_B, [\rho_x \otimes \mathbb{1}], [\mathbb{1} \otimes \sigma_y], [M_{c|z}]\}$. Here, we use the idea of Refs. [28–30] to assign to Π_A and Π_B the properties of a d-dimensional projector, i.e., we impose that $\Pi^2 = \Pi$ and $\mathrm{tr}(\Pi) = d$. Moreover, the states are fully contained in the support of these projectors, i.e., $\Pi_A \rho_x = \rho_x$ and $\Pi_B \sigma_y = \sigma_y$. We can also without loss of generality assume that the states are pure because the classical randomness in a mixed state can be absorbed into the global hidden variable λ. That ρ and σ live on separate spaces is captured through their commutation. Further, we assume the measurements to be projective, i.e., $M_{c|z} M_{c'|z} = \delta_{cc'} M_{c|z}$. These conditions together with the cyclicity of the trace imply many constraints on the moment matrix Γ. By construction, we have $\Gamma \succeq 0$ and the functional $\mathcal{R}_d$ can be expressed as a linear combination over the elements of Γ, provided the degree k of the monomials is sufficiently high. Thus, we can compute an upper bound on $\mathcal{R}_d$ as a semidefinite program. We have specifically selected monomials of length $k = 1$ supplemented with the length $k = 2$ monomials corresponding to the products $\{(\rho_x \otimes \mathbb{1}) M_{c|z}\}$.

However, the complexity of this semidefinite program scales rapidly with growing dimension, making it too costly for us to solve beyond the simplest relevant case of $d = 3$. This can be remedied by using symmetrization techniques to simplify the structure of Γ (see, e.g., [31–35]). For this, we observe that there exists local permutations of the variables of Alice, Bob, and Charlie that leave $\mathcal{R}_d$ invariant. By analyzing these permutations, one can reduce both the number of variables in Γ and its size. This is detailed in Supplemental Material [36]. Showcasing the relevance of this method, we have evaluated it for $d = 3, 5, 7$ and obtained the inequality (7). To obtain a certificate of the result, we have also derived the dual program (see Supplemental Material [36]) and confirmed that it gives the accurate result.

Violation with bound entanglement—We now show that the inequality (7) can be violated in the simplest case of $d = 3$ by using bound entanglement. For this purpose, we will consider a specific bound entangled (BE) state of two qutrits. Its density matrix takes the form

$$\rho_{\mathrm{BE}} = \begin{pmatrix} a_1 & 0 & 0 & 0 & a_2 & 0 & 0 & 0 & a_2^* \\ 0 & a_3 & 0 & 0 & 0 & a_4 & a_4^* & 0 & 0 \\ 0 & 0 & a_1 & a_5 & 0 & 0 & 0 & a_5^* & 0 \\ 0 & 0 & a_5^* & a_1 & 0 & 0 & 0 & a_5 & 0 \\ a_2^* & 0 & 0 & 0 & a_1 & 0 & 0 & 0 & a_2 \\ 0 & a_4^* & 0 & 0 & 0 & a_3 & a_4 & 0 & 0 \\ 0 & a_4 & 0 & 0 & 0 & a_4^* & a_3 & 0 & 0 \\ 0 & 0 & a_5 & a_5^* & 0 & 0 & 0 & a_1 & 0 \\ a_2 & 0 & 0 & 0 & a_2^* & 0 & 0 & 0 & a_1 \end{pmatrix}, \tag{8}$$

where (a_1, a_3) are real numbers and (a_2, a_4, a_5) are complex numbers. We select these numbers as

$$a_1 = \frac{4-\sqrt{3}}{27}, \quad a_2 = \frac{1-\sqrt{3}}{27} - \frac{i}{9\sqrt{3}}, \quad a_3 = \frac{1+2\sqrt{3}}{27},$$
$$a_4 = \frac{5-2\sqrt{3}}{54} + i\frac{5\sqrt{3}-12}{54}, \quad a_5 = \frac{\sqrt{3}-4}{54} + \frac{i}{18}. \tag{9}$$

Given these choices, one can easily verify that ρ_{BE} has unit trace and that it is positive semidefinite. Its rank is 7 and its nonzero eigenvalues with their corresponding multiplicities are $\{(\sqrt{3}-1)/3, (\frac{2}{9})^{\otimes 3}, [(2-\sqrt{3})/9]^{\otimes 3}\}$. We also see that ρ_{BE} has PPT. The spectrum of $\rho_{\mathrm{BE}}^{\mathrm{T}_A}$ is $\{[(4-\sqrt{3})/9]^{\otimes 3}, [(\sqrt{3}-1)/9]^{\otimes 3}, 0^{\otimes 3}\}$, which is nonnegative. The fact that ρ_{BE} is entangled can be certified from the computable cross norm or realignment criterion. To see this, let $\{\gamma_i\}_{i=1}^{d^2}$ be an orthonormal basis of d-dimensional Hermitian matrices, i.e., $\mathrm{tr}(\gamma_j \gamma_k) = \delta_{j,k}$.

The criterion states that for all separable states $\|C\|_{\mathrm{tr}} \equiv \mathrm{tr}\sqrt{C^\dagger C} \leq 1$, where $C_{j,k} = \mathrm{tr}(\gamma_j \otimes \gamma_k \rho)$ [37,38]. The violation obtained from choosing $\rho = \rho_{\mathrm{BE}}$ is $\|C\|_{\mathrm{tr}} = \frac{1}{3}(2\sqrt{3} + \sqrt{6} - \sqrt{2} - 1) \approx 1.166$.

Next, we use ρ_{BE} to violate the inequality (7). We select Alice's and Bob's encoding maps as unitary transformations. Specifically, we take $\Lambda_x(\tau) = U_x \tau U_x^\dagger$ with $U_x = X^{x_0} Z^{x_1}$, where $X = \sum_{k=0}^{d-1} |k+1\rangle\langle k|$ and $Z = \sum_{k=0}^{d-1} \omega^k |k\rangle\langle k|$ are the shift and clock operators, with $\omega = e^{(2\pi i)/d}$. We choose Bob's encoding maps as the same unitaries, namely $\Omega_y = \Lambda_y$. Note that we do not exploit the hidden variable and hence we drop the label for λ. Finally, we must select the measurements of Charlie. To this end, we define the bases

$$|e_{l|j}\rangle = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} \omega^{kl+jk^2} |k\rangle, \tag{10}$$

where $l \in [d]$ is the outcome label and $j \in [d]$ is the basis label. Select also $|e_{l|d}\rangle = |l\rangle$ (i.e., $j = d$) as the computational basis. For any prime d, the $d+1$ bases associated with j are mutually unbiased, i.e., it holds that $|\langle e_{l|j} | e_{l'|j'}\rangle|^2 = \frac{1}{d}$ when $j \neq j'$ [39]. For simplicity, we write $E_{l|j} = |e_{l|j}\rangle\langle e_{l|j}|$. Now, for the case of $d = 3$, we use these bases to construct the measurements of Charlie as

$$M_{c|z} = \sum_{b=0}^{d-1} E_{b|z} \otimes \tilde{E}_{b-c|z}. \tag{11}$$

Here, we have defined $\tilde{E}$ as a relabeling of the bases E and their eigenvectors. Specifically, we choose the tuple $(\tilde{E}_{0|z}, \tilde{E}_{1|z}, \tilde{E}_{2|z})$ as follows: $(E_{1|0}, E_{0|0}, E_{2|0})$ for $z = 0$, $(E_{0|2}, E_{2|2}, E_{1|2})$ for $z = 1$, $(E_{1|1}, E_{0|1}, E_{2|1})$ for $z = 2$, and $(E_{1|3}, E_{2|3}, E_{0|3})$ for $z = 3$. Evaluating the figure of merit, we obtain the value

$$\mathcal{R}_3 = \frac{1}{4} + \frac{1}{2\sqrt{3}} \approx 0.5387 > \frac{1}{2}. \tag{12}$$

This is a violation of the inequality (7). For completeness, we note that, as expected, the violation is much smaller than what is possible with a maximally entangled state. In that case, one can achieve $\mathcal{R}_d = 1$ for prime d. [23].

Maximally entangled states provide some insight as to why bound entanglement can achieve the reported violation (12). The connection stems from the observation that the ρ_{BE} is a Bell-diagonal state, i.e., that it can be written on the form $\rho_{\mathrm{BE}} = \sum_{k=1}^{9} q_k |\psi_k\rangle\langle\psi_k|$, where the mixing coefficients q_k correspond to the eigenvalues of ρ_{BE} and $\{|\psi_k\rangle\}$ is the Bell basis (i.e., the standard basis of maximally entangled two-qutrit states). To better understand the violation, we compute the value of $\mathcal{R}_3$ for each of the states $|\psi_k\rangle$ for the measurements that lead to the violation (12). Among the nine states, two give $\mathcal{R}_3 = 0$, three give $\mathcal{R}_3 = \frac{1}{4}$, another three give $\mathcal{R}_3 = \frac{1}{2}$, and the final one gives $\mathcal{R}_3 = \frac{3}{4}$. The first correspond to the null eigenvalues of ρ_{BE}, the second to the smallest nonzero eigenvalue, the third to the second largest eigenvalue, and the final to the largest eigenvalue. Thus, we see that the violation is made possible because of the skewed distribution of eigenvalues $\{q_k\}$, which is nevertheless compatible with PPT. It lets us assign only small (or even zero) eigenvalues to choices of $|\psi_k\rangle$ that achieve $\mathcal{R}_3 < \frac{1}{2}$, while it assigns a relatively large eigenvalue to the specific $|\psi_k\rangle$ that achieves a sizable violation. This leads to an average over the Bell basis, $\mathcal{R}_3 = \sum_k q_k \mathcal{R}_3(|\psi_k\rangle)$, which is sufficient to achieve (12).

Importantly, we also want to quantify how far the quantum correlations are from satisfying the inequality. This can be done in many different ways, but a common benchmark is to consider the robustness when the shared state is exposed to isotropic noise. This amounts to considering the density matrix $\rho_\nu = \frac{\nu}{d}\mathbb{1} + (1-\nu)\rho_{\mathrm{BE}}$, where $\nu \in [0,1]$ is the noise rate. Evaluating $\mathcal{R}_3$, we find $\mathcal{R}_3 = \frac{1}{12}(3 + 2\sqrt{3} - (2\sqrt{3} - 1)\nu)$. The critical ν for achieving a violation becomes

$$\nu = \frac{9 - 4\sqrt{3}}{11} \approx 0.1883. \tag{13}$$

This indicates a sizable robustness of the nonclassicality generated by the bound entangled state.

We remark that we have also performed a numerical search over bound entangled states, the encodings of Alice and Bob, and the measurements of Charlie, and we have found no larger violation of the inequality.

Higher dimensional violations—Given that large violations of classicality are possible in the prepare-and-measure scenario using three-dimensional bound entanglement, it appears natural to ask whether considering higher dimensions will lead to even stronger quantum correlations. This is *a priori* not obvious since several inequalities for the prepare-and-measure scenario are known to have diminishing quantum violations with growing dimension [40,41]. Interestingly, we find that for $d = 5$ and $d = 7$ even larger advantages are possible from bound entangled states than we found in dimension $d = 3$. To show this, we have employed an alternating convex search method [24] in which we set the encoding maps of Alice and Bob to the Weyl-Heisenberg unitaries, namely $\{X^j Z^k\}_{j,k=0}^{d-1}$, and consider optimization over the state and the measurements of Charlie. Each of these two optimizations can be written as a semidefinite program over matrices of dimension d^2, and we repeatedly iterate between the two in order to maximize the value of $\mathcal{R}_d$. We discuss this procedure in detail in Supplemental Material [36]. The results are given

TABLE I. Violations of inequality (7) using bound entangled states of dimension $d = 3, 5, 7$. For the optimal bound entangled state, we consider also its mixture with isotropic noise and determine the maximal noise rate for which a violation is possible.

	$\mathcal{R}_d$ w/o entanglement	$\mathcal{R}_d$ with bound entanglement	Isotropic noise tolerance
$d = 3$	1/2	0.5387	0.1883
$d = 5$	1/3	0.3862	0.2839
$d = 7$	1/4	0.2931	0.2869

in Table I. We see that the noise robustness improves significantly compared with the case of $d = 3$.

Conclusions—We have shown that bound entanglement is a resource for nonclassical correlations in the prepare-and-measure scenario, and that it can exhibit significant levels of noise tolerance. The noise tolerance exceeds that known for Bell nonlocality from bipartite bound entanglement by orders of magnitude. This noise robustness makes our schemes viable from an implementation perspective. Moreover, while most schemes in the entanglement-assisted prepare-and-measure scenario employ entangled measurements, ours uses only product measurements [42,43] to reveal the role of bound entanglement. This is many times a considerable simplification in the complexity of the measurement. Our results for $d = 3, 5, 7$ showcase a significantly increasing noise tolerance for quantum correlations from bound entanglement. It motivates a question for future work: can bound entanglement make possible a diverging correlation advantage? Specifically, in the limit of large dimension, can bound entanglement break the limitations of both classical models and quantum models without entanglement, when subject to noise rates that tend to one for large dimensions?

Code availability—The code used to prove the conjecture in (7) for $d = 3, 5, 7$ is available on GitHub [44].

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